Cognitive Ability Screen: guide for assessors

Version 3.1, October 2026. Use with `shared/Cognitive_Ability_Screen_Y11_Y12` (the student booklet, 14 pages). The screen is the same for Year 11 and Year 12 entrants. The grading sheet for the year applied for records the result.

What this screen is for, and what it is not. The screen gives the school a quick picture of how a student reasons with shapes, numbers and words, and how fast they work on a simple task. Its only purpose is to **plan learning support** once a student joins: which students may need help with English rather than with thinking, who may need extra time, where to look more closely. It is **never used to select**: no offer is made or refused on the strength of it, and it does not appear in the decision box of the final mark sheet. It is **not standardised**: the items were written by the school, there are no national norms, and the result is a set of **raw scores and the shape of the profile**, nothing more. It is **not a diagnosis** of any learning difficulty; that needs a specialist assessment. Until the school has its own reference, read every result cautiously. The school should **build its own reference once about 30 students have taken the screen** (Section 6.4).

Reusable: print once and keep in the assessor's folder. Do not write on these pages. Record the four raw scores and the profile note on the student's grading sheet.

1. At a glance

The screen comes after the second break, before the optional sciences. It can be taken in a small group with one invigilator. Timing is **strict**: every student gets exactly the same time, or the scores cannot be compared.

Part	What it measures	Time	Items and answers	What to watch
1	Non-verbal reasoning	**8 minutes**	12 items: pattern matrices, figure series, odd one out. Answer: a letter A to E.	No English needed beyond the instruction. The best guide to reasoning for a student with little English.
2	Quantitative reasoning	**8 minutes**	12 items: number series, number analogies, number matrices, simple relationships. Answer: a number, or a letter for item 6. No calculator.	Only simple arithmetic: it tests spotting the rule, not maths knowledge.
3	Verbal reasoning	**6 minutes**	12 items: word analogies, odd one out, simple deductions. Answer: a letter A to E.	English-dependent. Common words only, but a low score usually reflects English, not reasoning.
4	Processing speed	Practice untimed, then **90 seconds**	Symbol-digit coding: a key of 9 symbols, a practice row of 6, then 60 boxes. Score: correct boxes in 90 seconds, maximum 60.	Stop everyone at exactly 90 seconds. Check the practice row before the timed task.
	Total	**about 25 minutes** with instructions	Student booklet (14 pages); pencil and rubber; a timer that shows seconds	Record on the grading sheet: Non-verbal / 12, Quantitative / 12, Verbal / 12, Processing speed / 60, profile note.

You need: one student booklet per student, a pencil and rubber each, a stopwatch or a timer that shows seconds (a phone timer is fine), this guide, and the student's grading sheet. No calculators, no rulers needed, no dictionaries.

2. Before you start

Room. Quiet, bright, a clock the students can see, the booklet face down. The student writes their name on the cover. Students sitting the screen in a group should not be able to see each other's booklets.

Say (welcome): "This next part is a little different. It is a cognitive ability screen. It is not a test you can pass or fail, and it is not used to decide whether you get a place. It helps us plan how to support you when you join us. There are

four short parts, and each part is timed. Some questions are easy and some are hard. Nobody gets them all right. If you are not sure, choose the answer you think is best and go on. I will tell you when to start and when to stop each part. You may not use a calculator. Please do not turn any page until I say so. Open the booklet and read the first page with me." Read the Welcome and How to answer sections aloud, then the table of parts.

Access rules for the screen

Rule	What it means
Instructions	You may read the instructions and the worked examples of any part aloud in English, and repeat them as often as a student asks. You may read any item in Parts 1, 2 and 4 aloud. In **Part 3** read only the instructions and the examples, not the items: the words are the task.
Chinese	**Not used** by staff for any part of the screen, even where a Chinese-speaking colleague is present: the screen is not scored in the decision and nothing is lost by a student not understanding a word in Part 3, whereas a translation would make the verbal score unreadable. Do not mark L on this screen.
Calculator	**No calculator** in any part. The arithmetic in Part 2 is small on purpose.
Time	**No extra time** within the screen, even for a student with a known need for extra time in the papers: the time limits are part of what the screen measures. Instead, **note any known need** (extra time, a reader, dyslexia, ADHD, a visual or hearing difficulty, or anything the family has told you) on the grading sheet beside the profile note, so that the profile is read in that light.
Answers	Letters and numbers only, so English is a barrier in Part 3 alone. A crossed-out answer with a new one beside it is marked on the new one. Two answers left standing score 0.
Stopping	A student may put their pencil down early in any part and sit quietly; do not let them turn the page. Never push a distressed student: stop, take a break, and record it.

Do not explain an item, give hints, say whether an answer is right, or tell a student to hurry (except in the Part 4 script, which asks everyone to work fast). Say "thank you" rather than "good" or "right".

3. Giving the screen: the script and the timing

Keep the timer where you can see it. Start the timer the moment you say "Start" and say "Stop" when it reaches the limit, in the middle of a sentence if need be. Collect no booklets between parts: the student simply waits at the stop banner.

Part 1, Non-verbal reasoning (8 minutes). Say: "Turn to Part 1, Non-verbal reasoning, on page 2. Look at the example with me." Read the example aloud and point to the rule across the rows. **Say:** "There are three kinds of question." Read the three lines. **Say:** "You have 8 minutes for 12 questions. Write the letter in the box. If you are not sure, choose the best answer and go on. Stop when you reach the banner. Start." Time 8 minutes. Give a one-minute warning: "One minute left." At 8 minutes **say:** "Stop. Pencils down. Please wait."

Part 2, Quantitative reasoning (8 minutes). Say: "Turn to Part 2, Quantitative reasoning, on page 7. No calculator. Look at the three examples with me." Read the three examples aloud. **Say:** "In most questions you write a number. In question 6 you write a letter. You have 8 minutes for 12 questions. Start." Time 8 minutes, with a one-minute warning. At 8 minutes **say:** "Stop. Pencils down. Please wait."

Part 3, Verbal reasoning (6 minutes). Say: "Turn to Part 3, Verbal reasoning, on page 10. This part is about words. Look at the three examples with me." Read the three examples aloud. **Say:** "Write the letter in the box. If you do not know a word, choose the best answer and go on. I cannot explain any words in this part. You have 6 minutes for 12 questions. Start." Time 6 minutes, with a one-minute warning. At 6 minutes **say:** "Stop. Pencils down. Please wait."

Part 4, Processing speed (practice, then 90 seconds). Say: "Turn to Part 4, Processing speed, on page 13. Look at the key. Every symbol has a number: the circle is 1, the triangle is 2, the square is 3, the plus is 4, the cross is 5, the diamond is 6, the line is 7, the V is 8 and the T is 9. Under each symbol you write its number. Look at the example: a circle, so 1; a plus, so 4. Now do the practice row. Take your time. Use the key." Wait until every student has filled the six boxes (about 30 seconds). Check each practice row against the answers in Section 4.4; if a box is wrong, point to the symbol and the key and let the student correct it. Make sure every student has understood before the timed task. **Say:** "Now the timed task. Turn to page 14. The key is at the top of the page. When I say start, write the number for

each symbol. Go from left to right along each row, then start the next row. Do not skip any boxes. Work as fast as you can, but try not to make mistakes. You have 90 seconds. When I say stop, put your pencil down at once. Ready? Start." Time **exactly 90 seconds**; no warning. **Say:** "Stop. Pencils down." Do not let anyone finish the box they are on. **Say:** "Thank you. That is the end of the screen."

Timing in a group. Everyone starts and stops each part together. A student who arrives late sits the screen on another occasion: do not start them part-way through.

4. Answers

4.1 Part 1: Non-verbal reasoning

Mark the letter only. Each item has one correct answer; the rule is given so that you can check a disputed answer and so that a future author can see what each item tests.

Item	Type	Answer	The rule
1	Series	**C**	The number of dots goes up by one each time (1, 2, 3, 4), so the next figure has 5 dots.
2	Matrix	**B**	Each row has one shape; across the row there are 1, 2 then 3 of it. Row 3 needs three triangles.
3	Odd one out	**D**	In four figures the black dot is inside the shape. In D it is outside.
4	Series	**B**	The arrow turns 45 degrees clockwise each step (up, up-right, right, down-right), so the next points down.
5	Matrix	**A**	Each row and each column holds a circle, a square and a triangle once, and white, grey and black once. Row 3 lacks a square and white: a white square.
6	Odd one out	**E**	The number of dots equals the number of sides (triangle 3, square 4, pentagon 5, hexagon 6). E is a pentagon with 4 dots.
7	Series	**C**	Two things change: the black quarter moves one place clockwise each step (so it returns to the top left), and the lines below go up by one (1, 2, 3, 4, so 5). Only C has both.
8	Matrix	**D**	The third cell in each row holds all the lines of the first two cells added together (row 1: vertical plus horizontal; row 2: the two diagonals). Row 3: vertical and one diagonal, plus horizontal and the other diagonal, gives all four lines.
9	Odd one out	**A**	B, C, D and E are the same flag figure turned by 0, 90, 180 and 270 degrees. A is its mirror image: with the ball at the bottom, the flag points left instead of right.
10	Series	**E**	The shaded sectors go up by one each step (1, 2, 3, 4, so 5) and the small black square alternates top-left, bottom-right, so it is top-left next. Only E has 5 sectors and the square top-left.
11	Matrix	**B**	The third cell shows the small squares that are black in exactly one of the first two cells, not in both (row 1: top-left and top-right, with top-right, leaves top-left; row 2: top-left and bottom-left, with bottom-left and bottom-right, leaves top-left and bottom-right). Row 3: bottom-left is black in both, so the answer is top-left, top-right and bottom-right. A simple "add them" rule fails row 1, and "take away" fails row 2.
12	Matrix	**D**	Three rules at once. The outer shape and the inner shape each appear once in every row and column (a Latin square), and the inner shading goes white, grey, black across each row. Row 3 lacks a square outside and a triangle inside, and the third column is black: a square with a black triangle.

4.2 Part 2: Quantitative reasoning

Accept the number written in any clear form (for example 20 or 20.0). Working in the margin is fine and is not marked.

Item	Type	Answer	The rule
1	Series	**20**	Add 4 each time.
2	Series	**30**	Take away 5 each time.
3	Analogy	**5**	The second number is half the first (6 to 3, 8 to 4).

Item	Type	Answer	The rule
4	Matrix	**9**	In each row the first two numbers add to make the third (3 + 4 = 7, 6 + 2 = 8, 5 + 4 = 9).
5	Series	**32**	Double each time.
6	Relationship	**E**	2/3 is about 0.67, larger than 0.6, 3/5 (also 0.6), 0.55 and 1/2.
7	Series	**17**	The gaps grow by one: +1, +2, +3, +4, so +5.
8	Analogy	**13**	Double the first number and take away 1 (4 to 7, 10 to 19); 2 × 7 is 14, take away 1 gives 13.
9	Matrix	**21**	In each row the first two numbers multiply to make the third (2 × 3 = 6, 4 × 5 = 20, 3 × 7 = 21).
10	Relationship	**4**	Two circles and a triangle make 14, and one circle and a triangle make 9, so a circle is 5 and the triangle is 4.
11	Series	**38**	Two steps repeat: add 1, then double (3, 4, 8, 9, 18, 19), so the next is 19 × 2 = 38.
12	Analogy	**50**	Square the first number and add 1 (3 to 10, 5 to 26); 7 × 7 + 1 = 50.

In items 3, 8 and 12 two pairs are given so that only one rule fits both.

4.3 Part 3: Verbal reasoning

Item	Type	Answer	The reason
1	Analogy	**B**	Hot and cold are opposites; the opposite of big is small.
2	Odd one out	**C**	Four are fruits; a chair is not.
3	Analogy	**A**	A bird flies; a fish swims (what the animal does, not where it lives).
4	Deduction	**C**	Li Na is older than Tom, who is older than Mei, so Mei is the youngest.
5	Odd one out	**D**	Four are feelings; tall is not.
6	Analogy	**E**	A shoe is worn on the foot; a hat is worn on the head.
7	Deduction	**A**	All of Class 4 play football and Wang Hao is in Class 4, so he plays football. The other sentences may be true but do not have to be.
8	Odd one out	**D**	Four are lengths of time; a clock is an object.
9	Analogy	**B**	Begin and end are a pair of opposites in a process; ask and answer are the same kind of pair.
10	Deduction	**B**	Sofia is on Amir's left and Jun on his right, so Amir is in the middle.
11	Analogy	**A**	Always and never are opposites; the opposite of everything is nothing.
12	Deduction	**E**	If rain always stops the game and the game did not stop, it did not rain. A (it rained) is the common error: it reverses the rule.

The vocabulary is kept within the most common two thousand or so English words. Even so, a student new to English may not know "afraid", "angry", "begin" or "middle": that is the English barrier the profile is designed to show, not a fault in the student's reasoning.

4.4 Part 4: Processing speed

Practice row (not scored): square 3, circle 1, diamond 6, T 9, triangle 2, line 7.

The sixty boxes, row by row from the left:

Row 1	**2**	**9**	**8**	**9**	**3**	**8**	**7**	**5**	**6**	**4**	**5**	**9**
Row 2	**8**	**6**	**4**	**1**	**2**	**8**	**5**	**7**	**4**	**7**	**4**	**9**
Row 3	**8**	**6**	**3**	**1**	**6**	**1**	**7**	**3**	**7**	**2**	**7**	**5**
Row 4	**6**	**3**	**5**	**4**	**9**	**2**	**1**	**3**	**6**	**5**	**2**	**7**
Row 5	**4**	**9**	**1**	**9**	**3**	**5**	**8**	**6**	**8**	**1**	**2**	**3**

Key: circle 1, triangle 2, square 3, plus 4, cross 5, diamond 6, line 7, V 8, T 9. The symbols appear six or seven times each, and no symbol is ever next to itself.

5. Scoring

- **Parts 1 to 3:** one mark for each correct letter or number; no marks taken away for wrong answers. Raw score out of 12 for each part. Do not combine the three into a total: the parts measure different things and a total hides the profile.
- **Part 4, Processing speed:** the number of boxes **correctly** filled in 90 seconds, maximum 60. A wrong digit, an unreadable digit or an empty box scores 0 for that box; nothing is taken away. Count every box the student filled, including any filled out of order. Write the score as a number out of 60. Also note the number of **errors** (wrong digits) in the profile note if there are more than two: many errors with a high count suggests rushing; few boxes with no errors suggests a slow, careful worker.
- Write the four raw scores on the grading sheet as **Non-verbal / 12, Quantitative / 12, Verbal / 12, Processing speed / 60**. Do not convert them to bands, percentages or ages: there are no norms behind them.
- If a part was not given, or was stopped early, or a student arrived late, write "not given" or "stopped at item 7" rather than a score.

Rough expectations while the school has no reference. From the design of the items, a Year 11 or 12 entrant who reads English at the level the programme needs will usually score **8 or more** on each of Parts 1 to 3, and fill **35 to 50** boxes in Part 4. A Chinese-educated student with limited English will often score as well or better on Parts 1 and 2 and markedly lower on Part 3. These figures are a starting point only; replace them with the school's own reference (Section 6.4) as soon as it exists.

6. Reading the profile

6.1 What the shape means

Read the four scores together with the student's papers, the interview and what the family has told you. Look first at the **shape**: which parts are strong, which are weak **relative to the student's own other parts**, and whether the shape fits the rest of the day. A difference of **3 or more** between two of the parts out of 12 is worth noting; smaller differences are noise.

Profile	What it usually means	Provisional guidance for support planning
Non-verbal and quantitative both 9 or more, verbal low (4 or less, or 3 or more below the others)	Strong reasoning; English is the gap. The verbal items use common words only, so a low score here with strong non-verbal is almost always English, not reasoning.	Plan English support (ESL, reading and vocabulary) rather than learning support. Expect the student's results to rise quickly as their English grows. Say so in the readiness comment.

Profile	What it usually means	Provisional guidance for support planning
All three of Parts 1 to 3 at 4 or less	The student found reasoning with shapes, numbers and words all hard in the time given. This could be ability, anxiety, a misunderstanding of the task, a bad day or a known need.	Do not read it as a verdict. Check that the student understood the task (the practice and examples), look at the maths and science papers and the interview, and **discuss further assessment before a decision** (Admissions Guide Section 9). If the student joins, plan close monitoring in the first term and ask the family about any earlier assessment.
Processing speed well below the others (for example 25 or fewer boxes with Parts 1 to 3 at 8 or more, or a very slow rate with few errors)	The student reasons well but works slowly on a simple paper task. Common causes: reading and writing fluency, carefulness, unfamiliarity with the task, a visual or motor difficulty, or a processing speed difficulty.	**Consider extra time** in school assessments and check **reading fluency** (the sentence reading screen's fluency rating and the reading paper's stopping point). Watch for a student who runs out of time on every paper. If the pattern repeats in the first term, suggest a specialist assessment for access arrangements in public exams.
Strong profile, weak paper results (Parts 1 to 3 at 9 or more, but maths or science bands of 1 or 2, or English well below what the verbal score suggests)	The student can reason but has not learnt, or did not show, the content. Common causes: gaps in prior learning (a different school sequence, a change of school, time out of school), low motivation, or a day that went badly.	**Look for gaps in prior learning** (the NT topics on the grading sheet, the family's account of schooling) **or motivation** (the interview). Plan catch-up in the specific topics rather than general support, and set early targets to check that the student engages.
Quantitative strong, non-verbal weak (3 or more apart)	Comfortable with numbers, less so with visual patterns. Often a well-drilled maths background; sometimes a visual processing issue.	Usually no action. If the student also struggles with diagrams, graphs and geometry on the maths paper, note it for the maths teacher.
Non-verbal strong, quantitative weak (3 or more apart)	Sees patterns in shapes but not in numbers. Often number fluency or confidence; sometimes maths anxiety.	Compare with maths Section A (fluency). Plan number fluency support in maths if the paper agrees.
Even profile, all parts middling (5 to 8 each, Part 4 around 30 to 40)	A typical entrant. Nothing in the screen suggests a particular need.	No action from the screen. Plan from the papers and the interview as usual.
Everything high, Part 4 high (10 or more in each part, 45 or more boxes)	Fast, accurate reasoning across the board.	No support need from the screen. Consider stretch (for example Further Maths or an extra subject) if the papers agree.

6.2 What the screen cannot tell you

- It **cannot diagnose** dyslexia, dyscalculia, ADHD, a processing speed disorder or any other learning difficulty. It can only say "this pattern is worth a closer look". A **specialist assessment** by an educational psychologist or a qualified specialist teacher is needed for a diagnosis and for formal access arrangements in public examinations.
- It **cannot measure intelligence** or potential, and it must not be described to students or families as doing so. Describe it as "a short screen that helps us plan support".
- It **cannot separate** a student's reasoning from their English in Part 3, their attention on the day, their familiarity with this kind of task (students who have practised similar puzzles score higher) or their test anxiety. Part 1 is the most robust to English; none of the parts is robust to a bad day.
- It **cannot be compared with published tests**. A score of 9 out of 12 here means nothing in the terms of any commercial test, and the screen has not been checked for reliability or validity.
- A **single screen cannot track change**. If a student's profile raises a question, the answer comes from watching them learn in the first term, not from giving the screen again.

6.3 Talking to the family

Give the family the plain facts if they ask: the four raw scores, what each part is, and that the screen is for planning support and does not affect the decision. Do not offer an interpretation that sounds like a diagnosis. If the profile suggests a closer look (the second or third rows of the table in 6.1), say that the school would like to understand how the student learns best and ask whether any earlier assessment exists.

6.4 Building the school's own reference

The screen becomes more useful once the school knows what its own entrants typically score.

1. Record every student's four raw scores in the admissions tracker (the Year 11 and Year 12 sheets have columns for them), including students who were not offered a place.
2. After **about 30 students** have taken the screen, work out for each part the **median** and the **lower and upper quartiles** (the 25th and 75th percentile scores). Write them in a short note kept with this guide, with the date and the number of students.
3. From then on, describe each new student's score as "above the upper quartile", "in the middle half" or "below the lower quartile" of the school's own entrants, and use these in place of the fixed numbers in Section 5 and the table in 6.1 (for example "verbal below the lower quartile with non-verbal above the median" is the English-gap profile).
4. Review the reference each year, and separately for Year 11 and Year 12 entrants once each group has about 30 students. If an item is answered correctly by almost everyone or almost no one, or the same wrong answer is chosen by most students, note it in the version register so that the item can be reviewed at the next version.

The reference describes the school's own applicants, who are not a representative population: it supports comparison within the school and nothing more.

7. Recording the result

1. **Mark the booklet** with the answer tables in Section 4. Write the score for each part at its stop banner.
2. **Copy the four raw scores onto the student's grading sheet** (`Y11_Grading_Sheet_v3.1` or `Y12_Grading_Sheet_v3.1`) in the cognitive screen rows: **Non-verbal / 12, Quantitative / 12, Verbal / 12, Processing speed / 60**.
3. **Write the profile note** on the line provided: one or two short phrases, naming the shape and the provisional guidance from the table in 6.1, for example "Strong reasoning, English is the gap: ESL support" or "Slow coding (18) with good reasoning: check reading fluency, consider extra time" or "Even profile, no action". Add any known need or arrangement the family has told you about, and anything unusual on the day (anxiety, a misunderstanding of a task, a part not given or stopped early).
4. **Do not** enter the screen in the decision box of the final mark sheet. The final mark sheet carries the four scores and the profile note for the support plan only, marked "for planning support, never for selection".
5. **Enter the four scores in the tracker** so that the school's reference can be built (Section 6.4).
6. **File the marked booklet** with the student's other papers, or shred it under the school's policy.